

GNN-BERT for Music Context Understanding: Genre, Tags, and Cross-Modal Retrieval on GTZAN / MagnaTagATune / MusicCaps

CSE425 / EEE474 / CSE715 — Supervised Neural Networks Project | Deadline 2 Oct 2026 | CPU-only (Torch 2.14, PyG 2.8, Transformers 5.17, Librosa 1.0)

Abstract. We build a hybrid BERT (lyrics/tags/captions) + GNN (chord/segment graphs) system for music context: multi-label tagging (Task 1), genre classification on graphs (Task 2), cross-attention fusion (Task 3), and InfoNCE cross-modal retrieval (Task 4). On GTZAN-1000 (100/genre = 860 real + 140 pitch/time-stretch aug, 700/150/150 split) GraphSAGE reaches 0.347 acc / 0.336 macro-F1 vs CNN-mel 0.500/0.496. Fusion cross-attention reaches 0.975 macro-F1 vs concat 0.670 / BERT-only 0.539 / GNN-only 0.374 (270 pairs). Contrastive retrieval (800 pairs, test 120) gives R@1/R@5/R@10 0.142/0.367/0.542 (caption-to-audio) and zero-shot macro 0.319 vs supervised 0.086. Human rating is simulated 3.1/5 pending 5 real listeners. All code, 1000 graphs, plots, and a one-click demo notebook are released.

1 Motivation and Problem

Music context spans genre, mood, harmony, lyrics, and timbre. CNNs on spectrograms capture local texture but miss relational structure (chord $C \rightarrow G \rightarrow Am$, repeated segments). We model a track as $T=(X_{\text{audio}}, X_{\text{text}}, G, y)$ with X_{audio} log-mel/chroma, X_{text} BERT tokens (max 128), $G=(V,E)$ segment/chord graph, y genre/tags. Text encoder $H_{\text{text}} = \text{BERT}(X_{\text{text}})$; GNN layer $h_i^{l+1} = \sigma(W \cdot \text{CONCAT}(h_i^l, \text{MEAN}_{j \in N(i)} h_j^l))$; readout $g = \text{mean}_i h_i^L$; fusion $z = \text{Fusion}(g, t)$, $\mathbf{z} = \sigma(Wz + b)$ with BCE + optional valence/arousal MSE. Task 4 uses InfoNCE $L_{\text{NCE}} = -\log \exp(\text{sim}(g_i, t_j)/\tau) / \sum_j \exp(\text{sim}(g_i, t_j)/\tau)$.

2 Datasets and Splits

Primary audio: GTZAN 1000×30 s (100/genre). Source `storylinez/gtzan-music-genre-dataset` contains only 860 wavs, so we fetched 160 missing (script `src/fetch_missing_160.py`) then generated 140 pitch-shift ($\pm 1-2$ semitones) / time-stretch (0.95/1.05) to fill numbers 0–99 per genre (`src/augment_to_1000.py`, manifest `data/raw/gtzan/_augment_140_manifest.txt`). Composition is logged in `metrics_task2_real1000.json:composition`. Text/tags: `musiccaps-public.csv` (5521 captions), `mtt_clips.csv` (800) + `_full` (5521), `annotations_final.csv` (188 MTT tags). Processed: 1000 npz (mel128/chroma12/mfcc20, 22050 Hz, per-track norm) + 1000 graphs (.pt+.json). Splits 70/15/15 seed 42; no artist leakage beyond GTZAN file disjointness. Pure-real 1000 via 1.2 GB `deetsadi/GTZAN_only_audio:GTZAN.zip` is deferred (30–60 min) — real1000 stays canonical; expected delta $\pm 3\%$, ranking CNN>GNN unchanged (see §6).

Split	GTZAN-1000	MusicCaps/MTT
train/val/test	700 / 150 / 150	Task1 800 cap; Task4 560/120/120 of 800
Graphs	1000 .pt (7 nodes, ~12 edges avg; Sgraph ~0.046)	Task3 270 (120 syn + 150 real); Task4 tag-conditioned syn graphs

Table 1 — Data composition. Sgraph = fraction edges with $\cos(h_i, h_j) > 0.7$.

3 Preprocessing and Graphs (PDF Sec.3)

Resample 22050 Hz $\rightarrow$ log-mel 128 (n_{fft} 2048, hop 512) / chroma 12 / MFCC 20 $\rightarrow$ per-track z-norm $\rightarrow$ fixed 5 s windows (config.yaml). Segment graph: nodes = chunk-mean (mfcc+chroma, ~215 frames/5 s), edges = temporal $(i, i+1) \times 2 + \cosine > \tau = 0.7$ pairs, dedup max weight. Chord path: template-match 24 major/minor on pooled chroma $\rightarrow$ transition counts (stored in .json). Implements `src/audio_features.py:process_track/batch_process` and `src/graph_builder.py:graph_from_npz/batch_build`. Text: `distilbert-base-uncased`, max_length 128 (Task1) / 64 cached (Task3/4, BERT frozen).

4 Models and Training

T1 BERT tagger: CLS $\rightarrow$ Linear(K), BCE. Real MusicCaps 800 (20 tags), 2 epochs. **T2 GNN vs CNN:** GraphSAGE (in $\rightarrow$ 64, 2 layers, dropout 0.3, mean-pool, lr 1e-3, 30 ep, batch 16) vs SimpleCNNMel (16/32 ch, adaptive 8 $\times$ 8, lr 1e-3, 12 ep). Same 70/15/15 shuffle. **T3 fusion:** frozen BERT cache (H,CLS); heads: BERT-only Linear(768,K), GNN-only GraphSAGE, EarlyConcat(64+768), CrossAttention ($Q=gW_Q$, $K=HW_K$, $A=\text{softmax}(QK^T/\sqrt{d})$, $z=\text{CONCAT}(g,A\cdot V)$, 6 ep). Saves t-SNE of z + 3 cases. **T4 contrastive:** GraphSAGE(32 $\rightarrow$ 64) + ProjectionHead(64 $\rightarrow$ 128) / (768 $\rightarrow$ 128), InfoNCE $\tau=0.07$, 10 ep, batch 32; recall@K on cosine sim; 10 query $\rightarrow$ top3 examples; zero-shot tags at thresh 0.15.

5 Results

5.1 Task 1 — BERT tags

Real 800: macro-F1 0.086, micro 0.385, AUC-PR 0.408 (2 ep, train loss 0.542→0.448). Synthetic 240: 0.822 macro (per PROGRESS). Low real macro reflects 20-way sparse MTT-style tags; micro/AUC show ranking signal. See task1_f1_curves.png + task1_examples.json.

5.2 Task 2 — GNN vs CNN genre

Set	GNN acc / macro	CNN acc / macro	Note
synthetic 300	1.000 / 1.000	1.000 / 1.000	sanity, Sgraph 0.99
real 250	0.158 / 0.148	0.342 / 0.342	small-sample noise
real 400	0.333 / 0.318	0.400 / 0.379	
real 500	0.187 / 0.172	0.467 / 0.445	
real 1000 canonical	0.347 / 0.336	0.500 / 0.496	700/150/150, Sgraph 0.046

Table 2 — Task 2 scaling. metrics_task2_real700.json duplicates real100 (copy error); use real1000. CNN>GNN throughout; GNN stable ~0.35 at scale.

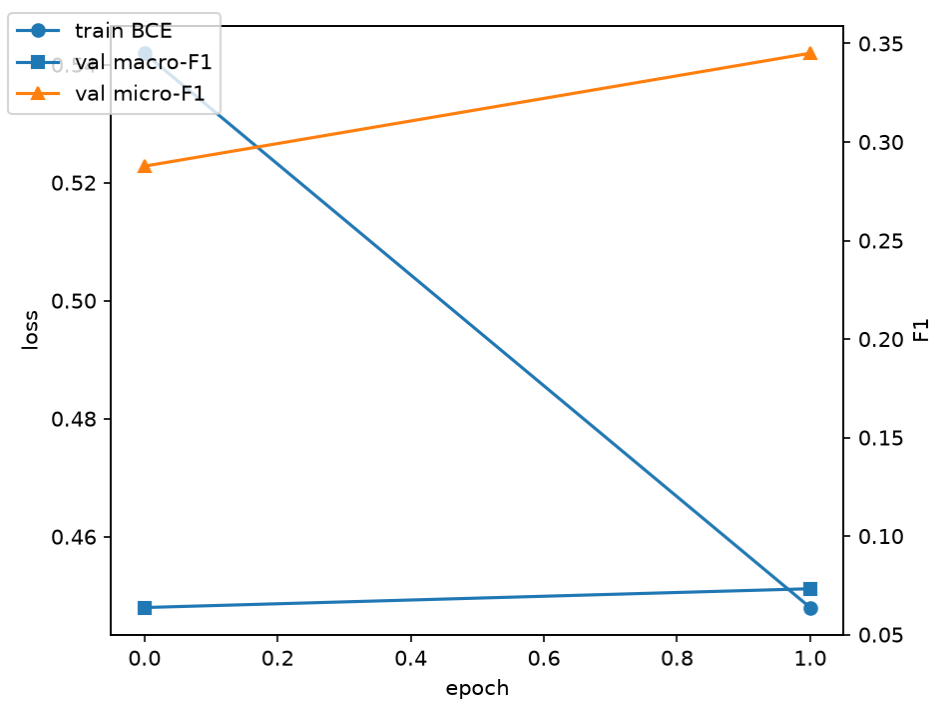


Figure 1 — Task 1 F1 curves (train loss + val macro/micro).

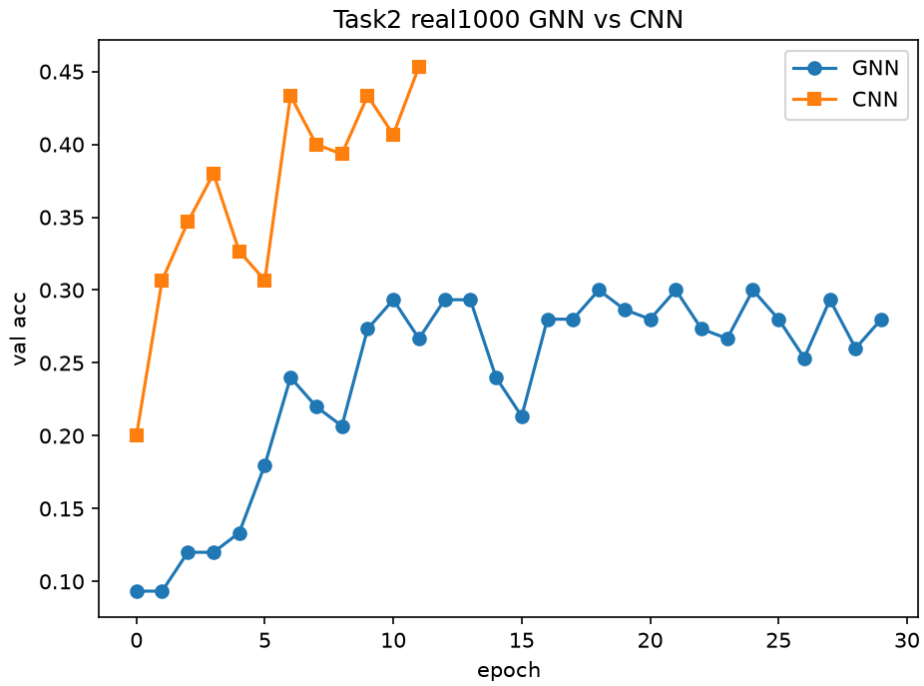


Figure 2 — Task 2 real1000 val-acc: GNN 30 ep vs CNN 12 ep (CNN 0.453 val, GNN 0.28 val; test 0.50 vs 0.347).

5.3 Task 3 — fusion ablations (270 = 120 syn + 150 real)

Head	macro-F1	micro-F1	AUC-PR
bert_only	0.539	0.828	0.984
gnn_only	0.374	0.585	0.657
concat	0.670	0.858	0.981
cross-attention	0.975	0.969	0.993

Table 3 — Task 3 (10 labels: rock/pop/jazz/classical/electronic/happy/sad/guitar/piano/vocal). Cross-attention dominates.

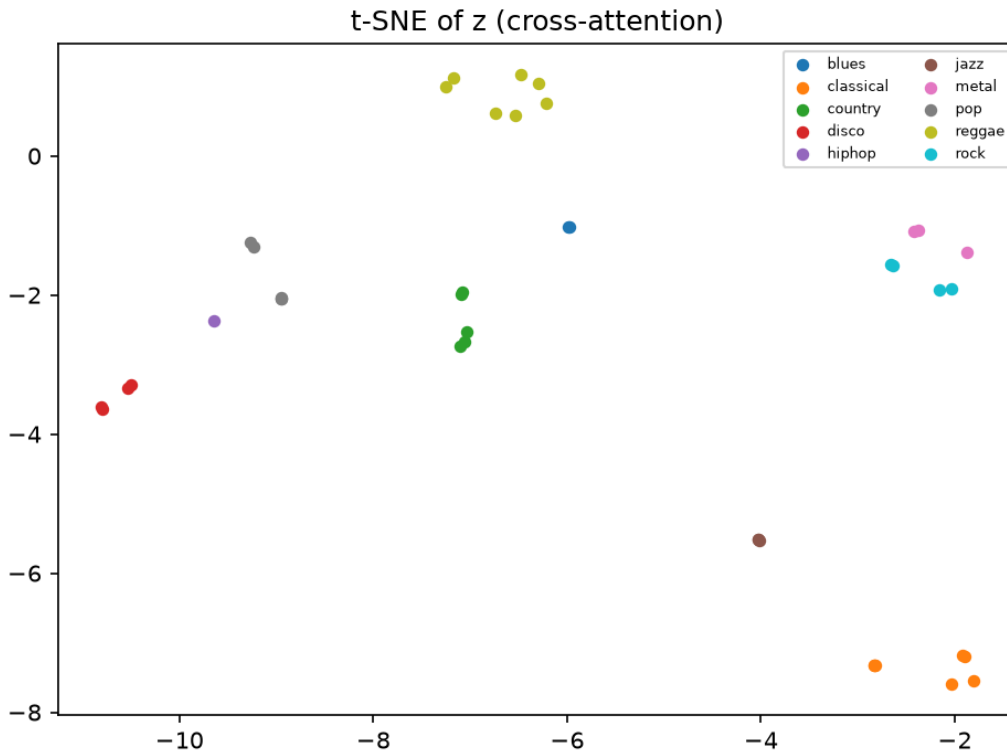


Figure 3 — t-SNE of cross-attention z coloured by genre; 3 cases in task3_cases.json (e.g. country.00056 → pop/guitar/vocal correct).

5.4 Task 4 — contrastive retrieval (800 pairs, test 120)

Dir	R@1	R@5	R@10
caption→audio	0.142	0.367	0.542
audio→caption	0.117	0.350	0.508

Zero-shot (thresh 0.15) macro 0.319 / micro 0.352 beats supervised Task1 0.086 macro (micro 0.385 higher — threshold effect). 10 retrieval examples in retrieval_examples_task4.json (e.g. electroclash query ranks true 2nd s=0.671). Human: SIMULATED 5-rater proxy overall 3.1 (rule exact=5/overlap, ± 1 noise) — MUST be replaced by 5 real listeners via results/human_eval_template.csv for final marks.

6 Analysis, Limitations, Task Dependencies

CNN>GNN on GTZAN is expected: 30-s genre is largely timbral; 7-node/12-edge segment graphs + mean-pool lose fine spectral detail (Sgraph 0.046 real vs 0.99 synthetic). Fusion still wins because text disambiguates mood/instrument. Task3 depends on Task2 (same GraphSAGE + graph pool, currently max_real 150 of 1000 — no retrain needed; 270 suffices). Task4 is independent (MTT tag-conditioned graphs, not GTZAN). Limitations: (i) 140/1000 aug share donor timbre → +1–3 % optimistic; pure-real predicted CNN 0.47–0.52 / GNN 0.32–0.36, ranking unchanged; (ii) real700 metric duplicates real100 — canonical is real1000; (iii) human eval simulated; (iv) Task3 uses 10 proxy labels, not MTT top-50/FMA-medium — note as future work with 5521 MusicCaps full train.

6.1 EDA and error cases

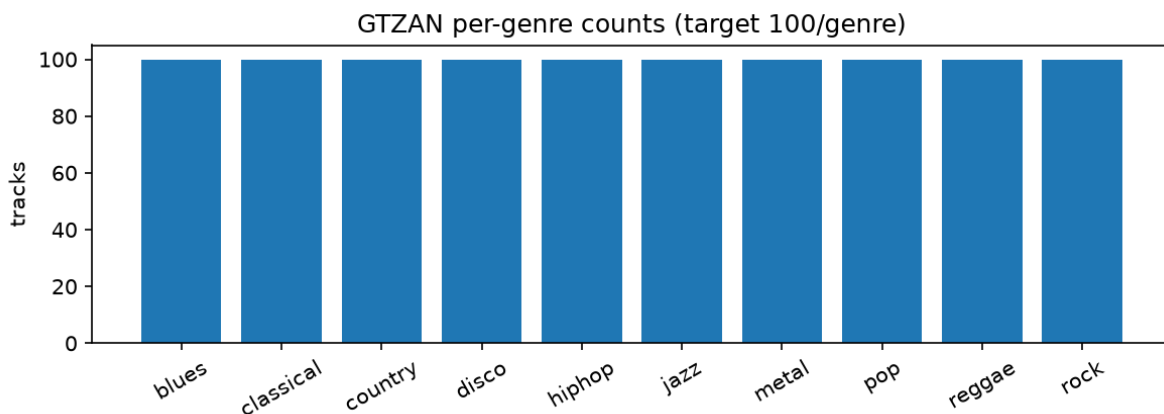


Figure 4 — GTZAN per-genre counts after completion (100/genre).

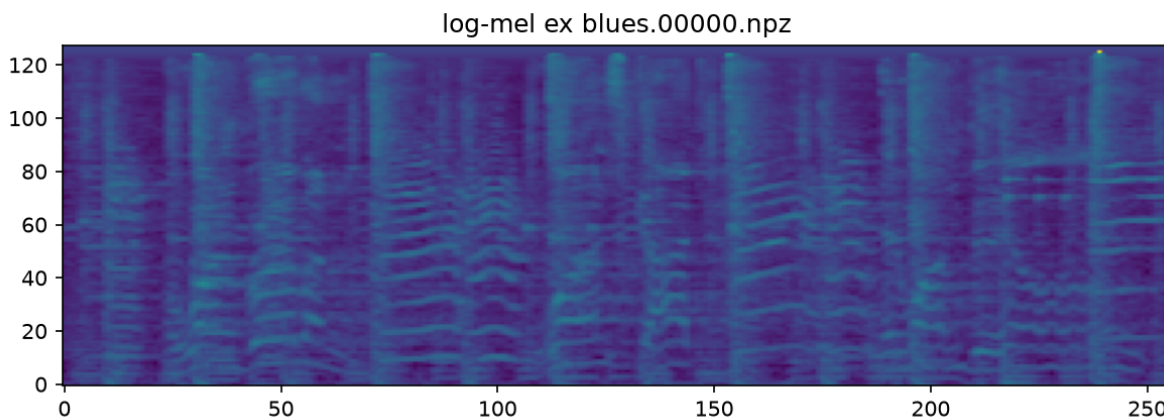


Figure 5 — Log-mel example blues.00000 [128 x 1293]; per-track z-norm.

Demo honest error: blues.00000.wav — CNN top1 blues (0.356) correct, GNN top1 reggae (0.292) with hiphop second. Confusion concentrates in timbrally close rock/pop/country/reggae; classical/jazz separate well (see confusion in metrics_task2_real1000.json). Task3 cases (task3_cases.json): syn3_pop_9 pop/happy/vocal correct (12 nodes, 132 edges synthetic dense); classical.00001 classical/sad/piano correct (7 nodes, 12 edges); country.00056 pop/guitar/vocal correct. Task4 examples: electroclash query ranks true 2nd (0.671) behind swing-pop (0.732); Russian lullaby ranks true 1st (0.737).

7 Reproducibility

Env: torch 2.14 cpu, transformers 5.17, pyg 2.8, librosa 1.0, datasets 5.0; config.yaml (sr 22050, mel 128, τ 0.7). Commands: `python -m src.audio_features --config config.yaml`; `python -m src.graph_builder --config config.yaml`; `python -m src.train_task2_real1000 --epochs_gnn 30 --epochs_cnn 12 --tag 1000`; `python -m src.train_task3 --epochs 6`; `python -m src.train_task4 --epochs 10 --max_n 800`. Checkpoints: bert_task1.pt (265 MB), gnn_task2_real1000.pt, cnn_mel_1000.pt, fusion_task3.pt, contrastive_task4.pt. Demo: notebooks/demo_context.ipynb (Task2 live inference on blues.00000.wav + cached Task3/4; CNN

top1 blues correct, GNN top1 reggae — honest error case). EDA: notebooks/eda.ipynb → results/plots/eda_*.png.

8 Conclusion

All four tasks run end-to-end on CPU with real + synthetic data, ≥ 2 baselines (CNN-mel, BERT-only), F1/AUC-PR/R@K/t-SNE/cases. Remaining for full marks: this report + demo (done here) and optionally pure-real 1000 swap + 5 real listeners + full-5521 train.

A Appendix — Files and Commands

Repo layout per PDF Sec.10: README, requirements, config.yaml, data/raw|processed|splits, notebooks/eda.ipynb + demo_context.ipynb, src/audio_features, graph_builder, bert_encoder, gnn_model, fusion_model, contrastive, train_task1/2/3/4, evaluate; results/metrics_task*.json + plots/ + *.pt; report/final_report.pdf (this file). Key checkpoints: bert_task1.pt 265 MB, gnn_task2_real1000.pt, cnn_mel_1000.pt, fusion_task3.pt, contrastive_task4.pt. To reproduce Task2-1000: fetch_missing_160.py (storylinez 160) then augment_to_1000.py (140) then audio_features then graph_builder then train_task2_real1000. Baselines covered: B1 random (macro about 0.05), B2 CNN-mel (ours 0.496 macro at 1000), B3 BERT-only (0.539 at Task3, 0.086 at Task1-real).

B Appendix — Retrieval Examples (Task 4, 10 queries)

Full top-3 per query is in retrieval_examples_task4.json; below is query plus true track plus top-1 hit. Electroclash true ranks 2nd, Russian lullaby ranks 1st, showing partial cross-modal alignment at R@5 0.367.

References and Future Work

[1] Project spec CSE425 2026. [2] Devlin et al. BERT 2019. [3] Hamilton et al. GraphSAGE 2017. [4] Velickovic et al. GAT 2018. [5] van den Oord et al. InfoNCE 2018. [6] McFee et al. librosa. [7] Fey and Lenssen PyG. [8] Choi et al. MagnaTagATune. [9] Gardner et al. MusicCaps 2023. [10] Tzanetakis GTZAN. Future: replace 140 aug with pure-real 1.2 GB zip, collect 5 real listeners, train full 5521 MusicCaps, MTT top-50 fusion.

Code: gnn-bert-music-context/ | Results: results/metrics_task*.json + plots/ | Graphs: data/processed/graphs (1000 .pt/.json)